Project Design Phase

Solution Architecture

Date	28 September 2026
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Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	

Solution Architecture

The architecture connects the passenger-facing booking experience with ServiceNow catalog configuration, validation rules, fare logic, QR code generation and the backend ticket and station data.

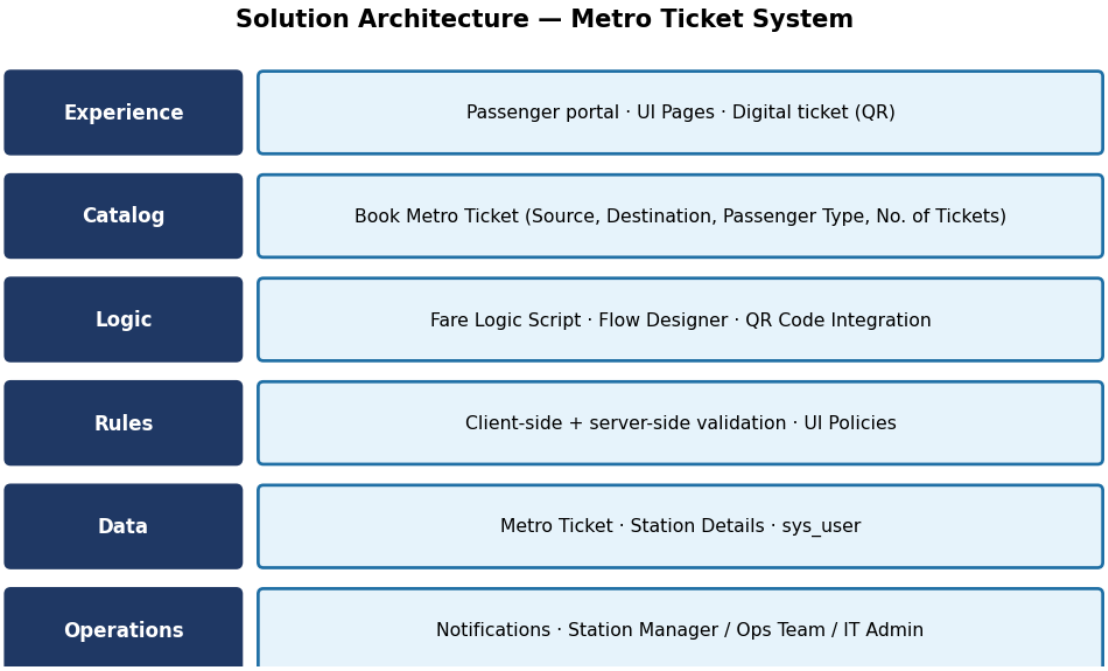


Figure 1: Metro Ticket solution architecture.

Layer	Components	Responsibility
Experience	UI Pages, passenger portal	Passenger UI.
Catalog	Book Metro Ticket item	Collect Source, Destination, Passenger Type, Number of Tickets.
Logic	Fare Logic Script, Flow Designer, QR Code Integration	Calculate fare and generate ticket.
Rules	Client/server validation, UI Policies	Validate and control fields.
Data	Metro Ticket, Station Details, sys_user	Persist records.
Operations	Notifications	Inform passengers and staff.

Data Architecture

Data architecture is the blueprint for how data is collected, stored, managed, integrated, and used across an organization. It defines the policies, standards, models, and rules that govern how data flows and supports business objectives.

Validation & Data Handling

Item	Description
Field validations	Non-empty source/destination, fare match, approved travel date.
Mandatory fields	Enforced on client and server side to prevent incomplete submissions.
Auto-populate logic	Default values (e.g., single, return journey) using catalog client scripts and UI policies.
Fields mapping	Process Automation Engine auto-maps fields and populates the values given by the user/clients.